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Electromyographic analysis of joint-dependent global synkinesis in the upper limb of healthy adults: Laterality of intensity and symmetry of spatial representation

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Abstract

The intensity and spatial representation of electromyographical (EMG) activity were examined to characterize the effects of limb dominance and movement direction upon global synkinesis (GS). Twenty-two healthy young subjects (11 men, 11 women) with a mean age of 24.7 years participated in this study. Three trials of EMG activities from eight primary muscles in the unexercised limb were recorded when a maximal isometric contraction in various directions was performed by the shoulder, elbow, and wrist of the dominant and non-dominant upper limbs. The features of GS, including intensity and spatial representation, were quantified with standardized net excitation levels (SNE) and relative excitation (RE), respectively. Our data indicated that (1) GS intensity was strongly limb-dependent with a larger SNE level arising when target joints of the non-dominant upper limb were active, (2) the GS intensity was more influenced by movement direction of the non-dominant limb than by that of the dominant limb, (3) the gradient change in GS intensity was observed bilaterally with a larger SNE level associated with contralateral movements of a proximal joint than a distal joint, and (4) GS spatial representations of the upper limbs were patterned and symmetrical, but seemly insensitive to movement direction. Laterality in GS intensity and structured GS spatial representation with symmetry could be a consequence of use-dependent hemispheric organization.
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Keywords: Synkinesis; Electromyography; Isometric contraction; Laterality

1. Introduction

Global synkinesis (GS), or contralateral irradiation, which has been defined as an unintended muscle contraction, occurs automatically with voluntary movement. This phenomenon can be readily observed in the homologous muscle of the opposite limb during unilateral movement [4], particularly for strenuous contraction against high resistance [39], difficult tasks in an unfamiliar environment [30], and for situations of neuromuscular fatigue [3]. A number of researchers have registered their concern about various characteristics of GS, relating them to general properties of the nervous system and functional specialization [21,28]. For instance, the frequent occurrence of GS has been reported for neurologically normal children [24] and aged people [35] due to immature myelination and degeneration of

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the transcallosal pathways. For patients suffering post-stroke hemiparesis, electromyographical data demonstrated that neurological impairments might result in stereotyped GS of the involved limb [9,10]. Although the precise mechanisms responsible for synkinesis are still disputable, many researchers favored decline in cortical inhibition from the contralateral motor cortex [16,33]. As intercortical effect could be crucial, debate as to whether laterality of the human hemisphere contributes to asymmetrical synkinesis invites researchers' attention. Many studies among healthy subjects have presented a greater amount or higher occurrence frequency of motor overflow when the non-dominant limb is active [5,12,38]. The invariant standpoint of GS asymmetry, however, was challenged by Parlow [30] who reported that synkinesis laterality depends primarily upon task characteristics. A greater overflow from the dominant to the non-dominant hands during finger-displacement tasks was noted, contrary to the typical asymmetry of GS pattern. Moreover, the extent of GS asymmetry may depend on how GS is characterized. It is noteworthy that earlier studies of GS symmetry was based merely upon occurrence frequency of the homologous muscle [5,30], overlooking non-homologous irradiation coupled with synergistic movement of the whole non-exercised limb [17,43]. Besides, several factors that may influence GS characteristics have not been systematically investigated. To the best of our knowledge, no work has been conducted to elucidate the association of synkinesis symmetry with joint involvement and movement direction for healthy subjects. We suspect that the intermanual asymmetry of contralateral irradiation could be task-dependent and subjective to various joint involvements as cortical representations and transcallosal connectivity are suggested to be joint-specific [40,42].

Using multi-channel surface electromyography, the objective of this study is to investigate intermanual asymmetry of GS patterns resulting from maximal isometric contraction of the contralateral upper limb in various movement directions. The intensity and spatial representation of the GS were both characterized, based upon the quantitative relations among irradiated EMG activity of different limb muscles. The finding of laterality of GS intensity and symmetry in GS spatial representation are of interest to investigators providing a better insight to cortical lateralization and interhemispheric communication.

2. Methods

2.1. Subjects

Twenty-two healthy young subjects (11 men, 11 women), with a mean age of 24.7 years (range = 20–38) featuring no history of any neuromuscular diseases were recruited in this study. These subjects were self-reported to be right-handed, and all revealed a normal full range of motion in the upper extremity. All study participants provided informed consent according to the guidelines set forth by the university concerning the protection of human subjects participating in trialing experiments.

2.2. Instrumentation

A Biodex dynamometer (Multi-Joint System 3 Pro, Shirley, NY, USA) was used to provide the resistance against isometric contraction in various directions. The real torque output of the dynamometer was connected to an oscilloscope (Model GOS-620, Instek, Malaysia) such that the subject could view his/her level of exertion in real-time and maintain the desired level continuously. Irradiated muscle activity contralateral to the exercised limb was recorded using preamplified bipolar surface electrodes (Iomed, Inc., USA; electrode spacing 2.5 cm; diameter 1.1 cm, and applying a gain of 380 and a common mode rejection ratio of 102 dB). Eight surface electromyographical (EMG) electrodes were applied over the bellies of the following muscles: biceps brachii (BI), brachioradialis (BRA), flexor carpi radialis (FCR), deltoid (DEL), triceps brachii (TRI), pronator teres (PRO), extensor carpi radialis (ECR), and pectoris major (PEC) [15]. The recorded myosignals were conditioned using analog low-pass filters (cut off frequencies set at 400 Hz), and were then digitized at 1 KHz using a computer program constructed on a Labview 6.0 platform (National Instruments, Austin, TX, USA).

2.3. Test administration

Those subjects who were not informed of the focus of our research interests, performed different-direction isometric contractions bilaterally using the shoulder, elbow, and wrist joints at maximal exertion levels, i.e., 100% of maximal voluntary contraction (MVC) level. In this study we defined the dominant condition was when target tasks were performed with the dominant limb, and vice versa. The order of the joint selected for testing, the particular limb selected, and the specific movement direction were randomized across subjects (Table 1). The MVC for a subject was determined when he/she gradually increased the force applied to a maximum value within 3 s commencement of muscle exertion and was able to sustain it for 1–2 s thereafter. There were a total of 10 isometric tasks selected for three tested joints for each limb. The position of the exercised upper extremity for performing isometric contractions with different joints was standardized according to the user's manual for the Biodex system. When performing required tasks

Table 1
Standard testing positions recommended by the Biodex system for voluntary isometric contraction of joints in the upper extremity in different directions

	Isometric task (seated)	Homologous muscle	Recommended testing position in the upper extremity (Biodex)
Shoulder	Abduction	DEL	Dynamometer orientation: 0° Dynamometer tilt: 10°
	Adduction	PEC	Seat orientation: 75° Seatback tilt: 70–85°
	Flexion	BI	Dynamometer orientation: 0° Dynamometer tilt: 0°
	Extension	TRI	Seat orientation: 15° Seatback tilt: 70–85°
Elbow	Flexion	BI and BRA	Dynamometer orientation: 30° Dynamometer tilt: 0°
	Extension	TRI	Seat orientation: 0° Seatback tilt: 85°
Forearm	Pronation	PRO	Dynamometer orientation: 0° Dynamometer tilt: 5° (shaft down)
	Supination	BI	Seat orientation: 90° Seatback tilt: 85° Elbow flexion: 90°
Wrist	Flexion	FCR	Dynamometer orientation: 0° Dynamometer tilt: 0°
	Extension	ECR	Seat orientation: 0° Seatback tilt: 85° Elbow flexion: 90°

DEL = deltoid; BRA = brachioradialis; BI = biceps brachii; FCR = flexor carpi radialis; PEC = pectoris major; TRI = triceps brachii; PRO = pronator teres; ECR = extensor carpi radialis.

with the working joint, all subjects in the test were secured in the dynamometer chair with the hips, knees, and ankles stabilized roughly at 90° flexion, whilst the unexercised upper limb was kept relaxed. The subject was strongly motivated to perform isometric contraction of 100% MVC for a period of 5 s following the investigator's verbal encouragement, and then the "mirrored" EMG from the unexercised limb was recorded. The subject's appropriate exertion level was ensured by the individual being able to visualize, in real-time, a graphical representation of his/her activity level, and match a line for true dynamometer torque output with a line representing the required level as displayed on the oscilloscope connected to the dynamometer. Each subject performed the isometric task three times with rest intervals during trials for a period of 2 min. Between rest intervals, a 5-s background activity level was carefully monitored and recorded, when the subject was reminded to relax maximally. When the whole session of experimentation was completed for one limb, a similar protocol was repeated for the corresponding limb 10 min later.

2.4. Data analysis

Off-line EMG analyses included the removal of linear trend in the raw EMG data and possible data artifacts using a 10th-order digital Butterworth bandpass filter (cut off frequency = 40–400 Hz). Root mean square data (RMS) were determined following EMG conditioning, and the RMS value for three trials was averaged before subtraction of background activity to obtain net excitation of an irradiated muscle. Normalized with EMG background activity, the summation of the net excitation from all recorded muscles resulted in a standardized net excitation (SNE), which was taken as a fair representation of GS intensity. Specifically, relative excitation (RE) was defined as comparative contribution of the amount of irradiation from a given muscle to overall SNE value [22]. The quantitative relationship of RE among recorded muscles denotes GS spatial representation. The selection of EMG normalization with SNE rather than a percentage of maximal EMG value contralateral to the

muscle that performs the target task is because completion of a target task generally needs coactivation of many functional synergists. Normalization with maximal EMG value from a single muscle was not likely appropriate.

Repeated-measures two-way ANOVA tests were used to examine (1) any difference in EMG background activity at rest with respect to limb dominance and muscle, and (2) the effects of limb dominance and movement direction of a target joint upon SNE. For both dominant and non-dominant conditions, the independence of RE of each muscle among various involved joints was examined using repeated-measures one-way ANOVA testing. The level of significance for each ANOVA, and post-hoc testing was set to alpha = .05. Signal processing and statistical analyses were completed using Matlab v. 6.0 (Mathworks Inc., Natick, MA, USA) and SPSS for Windows v.10.0 (SPSS Inc., Chicago, IL, USA).

3. Results

The background activity at rest was well controlled in support of ANOVA statistics. The result indicated that the background activity did not differ between the dominant and non-dominant limbs [$F(1,21) = 1.88$, $p = .185$] as well as among muscles [$F(7,147) = 1.57$, $p = .147$]. Fig. 1(a) and (b) reveals the EMG data recorded from a typical subject during isometric shoulder abduction for both the non-dominant limb and the dominant limb, respectively. It was apparent that irradiation was spread over several muscles in the unexercised limb during shoulder abduction without it being confined exclusively to the contralateral homologous DEL. Such an observation validated the use of SNE to characterize GS intensity inclusive of non-homologous irradiation.

Fig. 2(a)–(c) reveals the mean and standard deviation of SNE as a function of whether joint movements of the dominant (right) or non-dominant (left) limbs were performed. For the shoulder joint (Fig. 2(a)), repeated-measures ANOVA suggested that there was a significant difference in the SNE value for limb dominance and movement directions featuring a significant interaction effect [$F_{\text{limb}}(1,21) = 6.69$, $p = .017$; $F_{\text{direction}}(3,63) = 9.33$, $p < .001$; $F_{\text{limb}\times\text{direction}}(3,63) = 1.34$, $p = .042$]. Post-hoc analysis suggested that movement direction affected SNE significantly for the non-dominant and dominant conditions [$F_{\text{ND}}(3,63) = 8.29$, $p < .001$; $F_{\text{D}}(3,63) = 3.03$, $p = .036$]. This SNE difference between non-dominant and dominant conditions was evident for shoulder abduction [$F(1,21) = 6.26$, $p = .021$].

For the elbow joint (Fig. 2(b)), differences in the SNE were significant between the dominant and non-dominant conditions and among movement directions

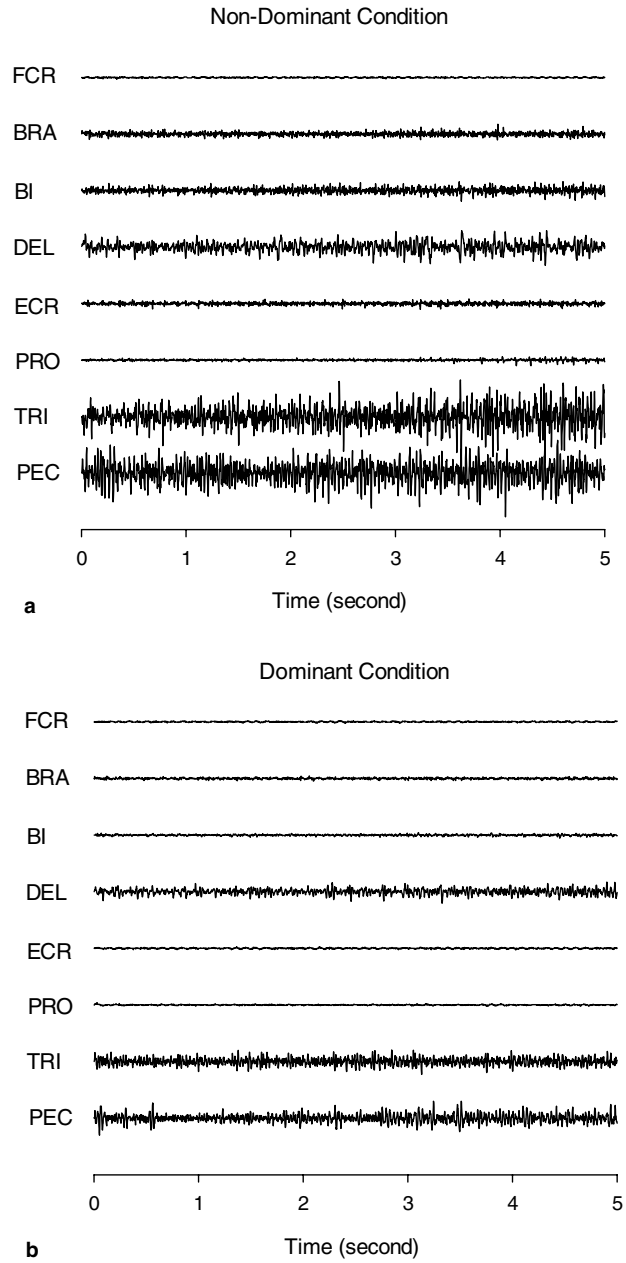
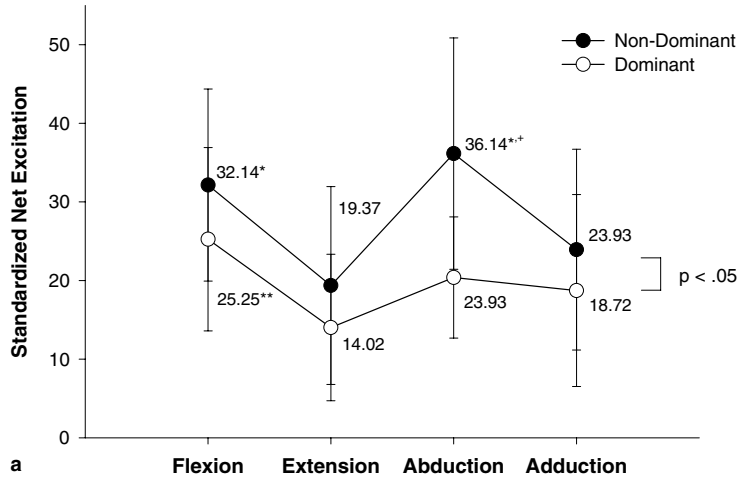
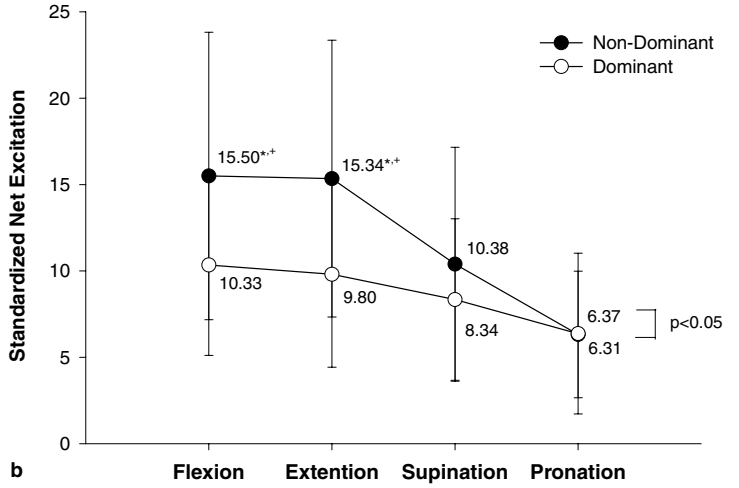


Fig. 1. A typical example of irradiated muscle activity in the contralateral upper limb associated with isometric shoulder abduction at 100% MVC: (a) when shoulder abduction of the non-dominant limb was performed, (b) when shoulder abduction of the dominant limb was performed.

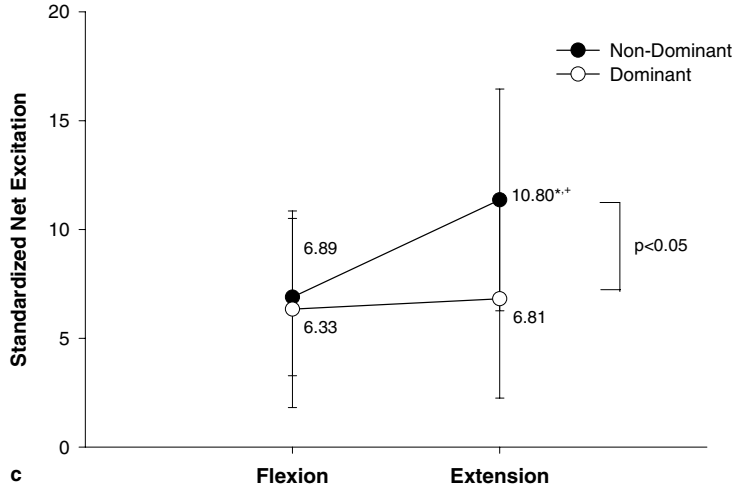
[$F_{\text{limb}}(1,21) = 5.28$, $p < .032$; $F_{\text{direction}}(3,63) = 9.74$, $p < .001$; $F_{\text{limb}\times\text{direction}}(3,63) = 1.73$, $p = .171$]. Non-dominant elbow movements generally introduced greater SNE levels, especially for elbow extension and flexion [$F_{\text{ext}}(1,21) = 5.17$, $p = .034$; $F_{\text{flex}}(1,21) = 4.17$, $p = .048$] than was the cases for supination and pronation. The effect of movement direction of the elbow joint upon SNE levels was clearly observable only for the non-dominant condition, but was not significant



SNEflex, SNEabd are significantly larger than SNEext, SNEadd in the non-dominant condition (*: p < .05)
SNEflex are significantly larger than SNEext in the dominant condition (**: p < .05)
SNEabd in the non-dominant condition are significantly larger than that in the dominant condition; (+: p < .05)



SNEflex, SNEext are larger than SNEsup, SNEpro in the non-dominant condition (*: p < .05)
SNEflex, SNEext in the non-dominant condition are larger than those in the dominant condition (+: p < .05)



SNEext is significantly larger than SNEflex in the non-dominant condition (*: p < .05)
SNEext in the non-dominant condition is significantly larger than SNEext in the dominant condition (+: p < .05)

Fig. 2. A graphical summary of the standardized net excitation level resulting from isometric tasks conducted in different directions: (a) shoulder; (b) elbow; (c) wrist (dominant: –●–, non-dominant: –○–).

for the dominant condition [$F_{ND}(3,63) = 5.65$, $p = .002$; $F_D(3,63) = 1.18$, $p = .323$].

For the wrist joint (Fig. 2(c)), SNE was dependent upon both limb dominance [$F_{limb}(1,21) = 9.36$, $p = .006$] and movement direction [$F_{direction}(1,21) = 4.61$, $p = .044$] without a significant interaction effect noted [$F_{limb\times direction}(1,21) = 3.82$, $p = .064$]. Non-dominant wrist movements again elicited a more remarkable level of SNE than dominant wrist movements, but this laterality was significant only for wrist extension [$F_{ext}(1,21) = 14.83$, $p = .001$; $F_{flex}(1,21) = .23$, $p = .640$]. In spite of a significant effect of movement direction upon GS intensity, post-hoc analysis revealed that only non-dominant wrist extension brought about a greater degree of SNE than non-dominant wrist flexion [$F(1,21) = 7.51$, $p = .012$], whilst SNE proved to be independent of movement direction when the dominant wrist was active [$F(1,21) = .09$, $p = .770$].

The comparison of the joint effect upon GS intensity during isometric flexion/extension. For flexion movement, ANOVA statistics indicated that there was a clear topological tendency of SNE difference between proximal and distal joints [$F_{ND}(2,42) = 41.35$, $p < .001$; $F_D(2,42) = 25.92$, $p < .001$], and the SNE value for shoulder flexion of both limbs was, respectively, larger than the corresponding value for wrist flexion ($p < .005$). On the other hand, when dominant isometric extension was performed, the joint effect upon SNE was

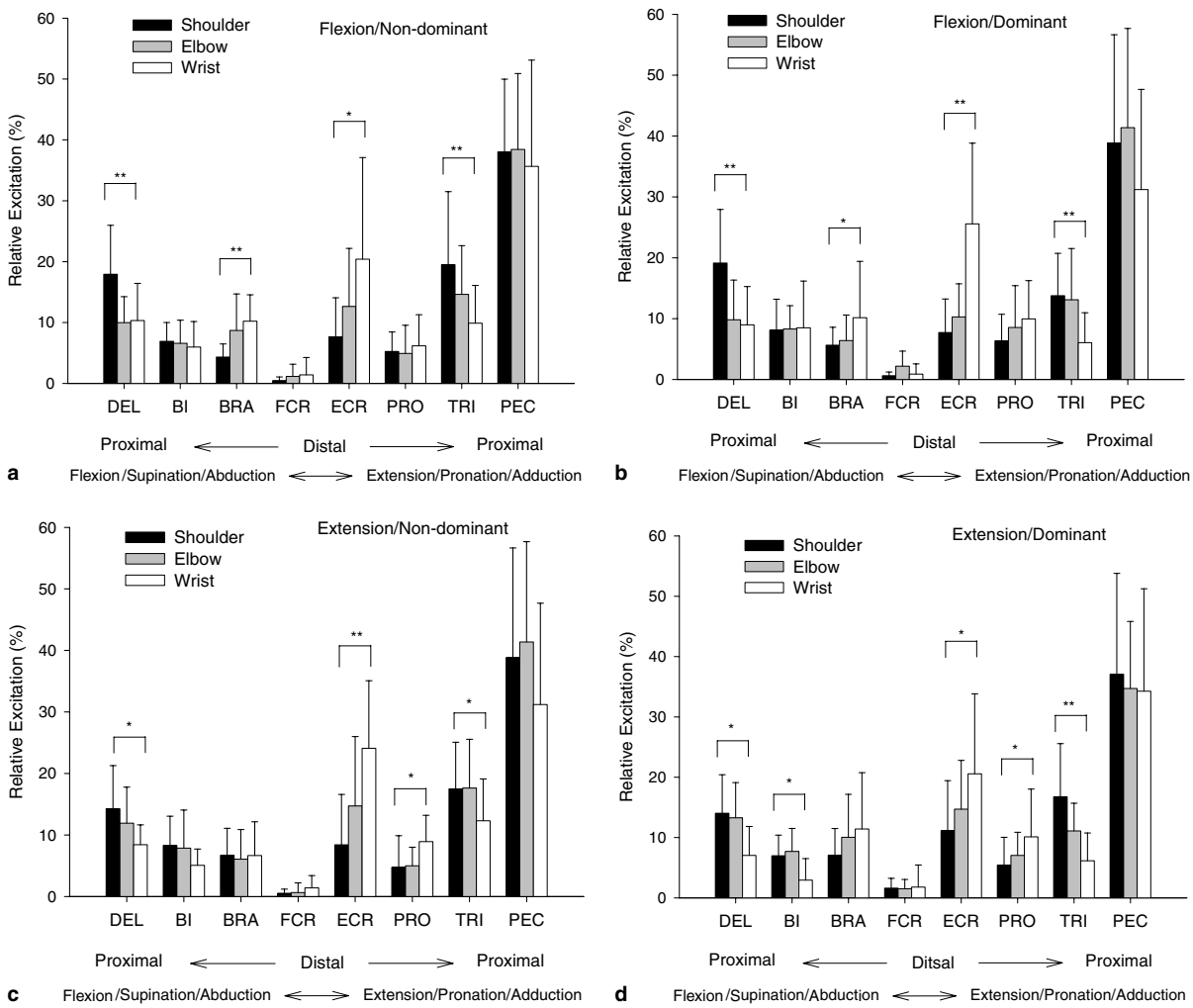


Fig. 3. Schematic expressions of the relative contribution of each irradiated muscle to global synkinesis: (a) and (b) isometric flexion contraction of the shoulder, elbow and wrist joints in the non-dominant and dominant conditions; (c) and (d) isometric extension contraction of the shoulder, elbow and wrist joints in the non-dominant and dominant conditions. Fig. 3(a)–(d) show the effect on RE which is symmetrical for the non-dominant and dominant conditions. (e) and (f) isometric abduction/adduction contractions of the shoulder joint in the non-dominant and dominant conditions; (c) and (d) isometric pronation/supination contractions of the elbow joint in the non-dominant and dominant conditions. Fig. 3(e)–(h) are examples that illustrate no observable difference on RE with respect to movement direction for the non-dominant and dominant conditions. Notice that RE is rather symmetrical for a given contralateral movement. (Significant difference in relative excitation level among shoulder, elbow, and wrist joints, *: $p < .05$, **: $p < .005$.)

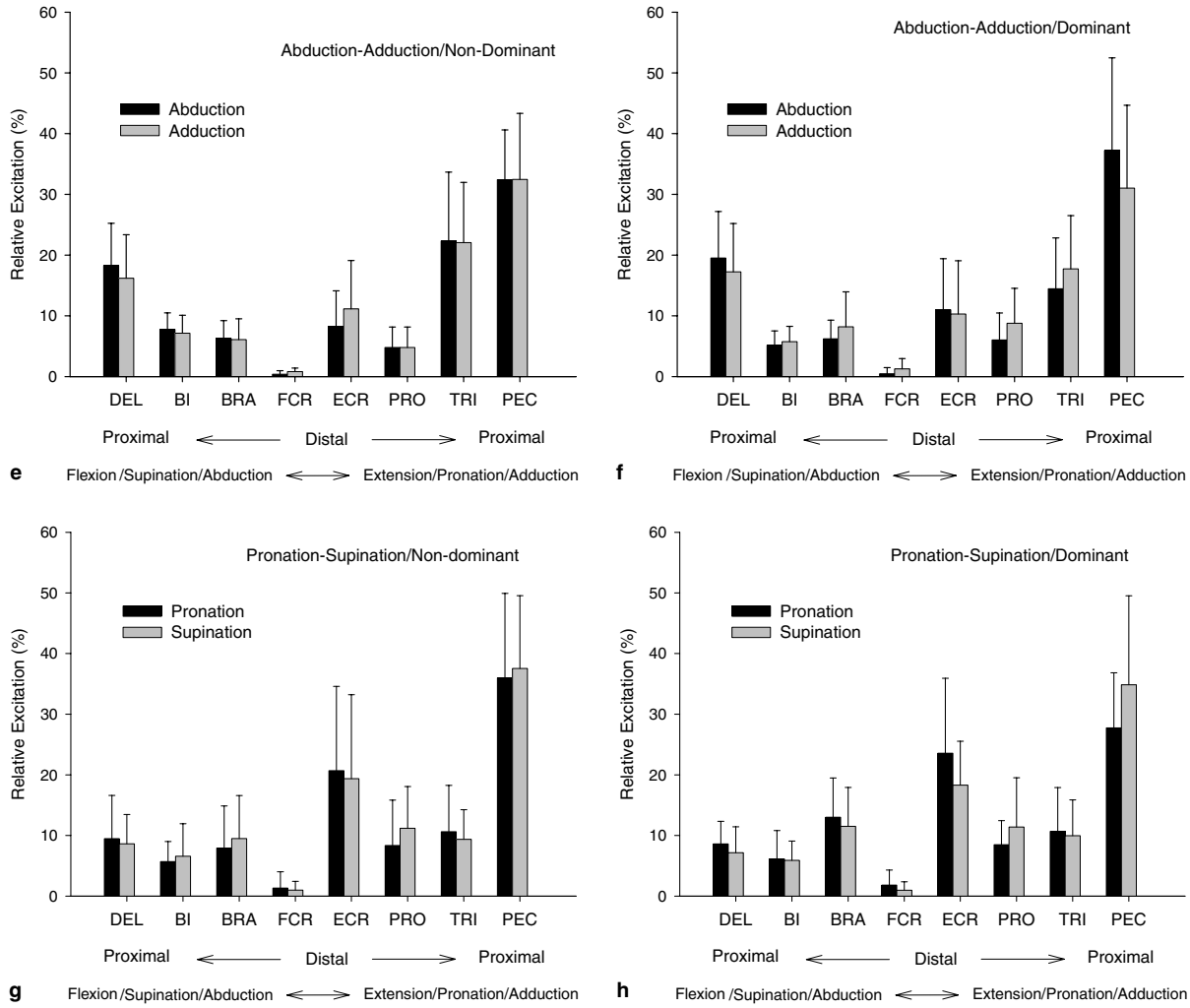


Fig. 3 (*continued*)

marginally significant [$F_{\text{joint}}(2,42) = 2.93$, $p = .065$]. The SNE value for the isometric extension of the non-dominant joint, however, varied considerably for the involved joint, with a greater SNE value associated with extension of the proximal joint relative to distal-joint extension [$F(2,42) = 4.01$, $p = .025$].

Interestingly, in addition to GS intensity, spatial representation of GS was subjective to an intricate joint effect when the dominant or non-dominant limb was active. Fig. 3(a)–(d) depicts the RE level of the eight irradiated muscles contralateral to the exercised joints, in the directions of flexion and extension. It was evident that the spatial representations of GS in terms of the RE were structured. For both the dominant and non-dominant conditions, one-way repeated ANOVA tests suggested that contralateral shoulder movements tended to bring about greater relative excitation of the proximal muscles (DEL or TRI) than the corresponding distal muscles, whilst relative excitation of the distal muscles (ECR, PRO, or BRA) was relatively enhanced by contralateral wrist movements ($p < .05$). In spite of the joint effect on spatial representation of the GS, for all target tasks conducted, the RE of the synergistic muscles responsible for extension/adduction were relatively larger as a consequence of contralateral movements than the RE of the muscles responsible for flexion/abduction of the same joint. Fig. 3(e)–(h) shows spatial representation of shoulder abduction/adduction and elbow pronation/supination in non-dominant and dominant conditions. It was clear that the effect of movement direction of a joint upon GS was inconsequential. Also, the particular topological-relevant spatial pattern during contralateral flexion and extension was applicable to GS during contralateral elbow pronation/supination and shoulder abduction/adduction. Collectively, spatial representation of GS for the dominant and non-dominant conditions appeared to be symmetrical, governed by a similar joint effect with greater relative excitation attributable to synergists in charge of extension and adduction.

4. Discussion

There were several novel findings resulting from this study. GS intensity was dependent upon both involved joint and movement direction when the non-dominant limb was active, while GS intensity was affected, to a much lesser extent, by involved joint and movement direction for the dominant condition. By contrast, the spatial representation of GS activity was similar for both dominant and non-dominant conditions independent of movement direction.

4.1. Directional-dependent laterality of GS intensity

When target joints of the non-dominant limb performed isometric contractions, a more remarkable GS intensity developed relative to that triggered by the comparative joints of the dominant limb. Consistent with previous studies based on occurrence frequency of mirror movement [6,25,30], our EMG finding accounting for non-homologous irradiation agreed with laterality of GS intensity during strenuous motor tasks. Moreover, laterality of GS intensity was directionally sensitive so that the difference in the amount of contralateral irradiation became marked specifically during flexion/abduction of the shoulder, flexion/extension of the elbow, and extension of the wrist (Fig. 2(a)–(c)).

The asymmetry of GS intensity may be of spinal or central origins [37,43]. For the upper limb, contralateral effects at cortical level may be much more influential than at segmental level. Many researchers have demonstrated that the H reflex of the FCR and F wave of the abductor pollicis brevis were not potentiated during moderate and strong voluntary movements of the contralateral upper limb [20,29], whilst motor-evoked potentials in the adductor pollicis and elbow flexors elicited by cortical stimulation increased in size [29,37]. As greater exertion yields stronger irradiation for additional activation of cortical neurons [37], it seemed reasonable to relate asymmetry of GS intensity with the strength difference that the dominant and non-dominant limbs were able to develop. However, we challenge this strength-dependent assumption for laterality of GS. First, torque generation by the dominant limb was generally larger than by the non-dominant limb in the tests. Next, a strength-dependent assumption can not account for the directional effect on GS intensity. For instance, the peak torque required for shoulder adduction is almost triple that required for shoulder flexion [11], but shoulder flexion invoked substantially greater GS intensity than was the case for shoulder adduction, particularly when isometric contraction of the non-dominant shoulder was performed (Fig. 2(a)).

On the other hand, the association between laterality of GS intensity and target movement may be attributable to hemispheric specialization [30]. Todor and Lazarus [39] have proposed that differential use of the dominant limb refines neural connectivity of the dominant hemisphere [38], involving a training-induced reduction in transcallosal inhibition and adaptation in cortical plasticity [14,31]. With conditioning transcranial magnetic stimulation, Ridding et al. [31] discovered a significant change in evoked potential of the dorsal interosseous muscle for professional musicians, and the authors referred such an alternation to train-induced asymmetry of transcallosal connectivity with a reduced inhibition of the dominant hemisphere upon the non-dominant hemisphere. Hence, when target movements of the non-dominant limb were performed, GS in the dominant limb conveyed primarily by the ipsilateral corticospinal tract might be suppressed to a lesser extent because of asymmetry of transcallosal inhibition [13,31]. As a directional effect on GS intensity was evident particularly for the non-dominant condition, we further hypothesize that primary motor cortex cells in the non-dominant hemisphere, especially those encoded with movement force and direction [23,34], were selectively disinhibited by the reduction of transcallosal inhibition from the dominant hemisphere.

Certainly, the hypothetical linkage between GS intensity and the degree of directional tuning of motor cortical activity discussed herein should be tested in the future. In contrast to a directional effect on asymmetry GS intensity, a progressive decline in GS intensity from proximal to distal joints was still observed for both limbs (Fig. 3(a) and (b)). The fact indicated that the joint effect on GS intensity was preserved bilaterally, although the gradient change in GS intensity was marginally evident for isometric extension in the dominant condition. Acknowledging that the proximal joints generate greater torque than distal joints [1,11], the symmetrical joint effect on GS intensity conformed to the strength-dependent assumption.

4.2. Structured spatial representation of GS with symmetry

In spite of laterality in GS intensity, the GS spatial representation during flexion and extension movements for dominant and non-dominant conditions was symmetrical, sharing some common features independently of movement direction. First, joint movements always yielded a greater amount of excitation in the extension/adduction synergists of a joint (PEC, TRI, ECR) as compared to that in flexor/abduction synergists (DEL, BI, FCR); Next, there was a universal spatial relationship among RE for both extensor/adductor and flexor/abductor synergist pairs for all target movements (extensor/adductor: PEC > TRI, PRO, ECR; flexor/abductor: DEL > BI, BRA, FCR). Third, irradi-

ation in distal extensor (ECR) was relatively enhanced by contralateral distal movements that associated with corresponding decline in RE of proximal DEL and TRI (Fig. 3(a)–(d)). For symmetrical spatial representation of GS, there was probably a conceptual center residing in both hemispheres to modularize the spatial relationship among several irradiated muscles. Physiologically, preferential proximal irradiation was compatible with anatomical findings in monkeys that motoneurons innervating the axial and shoulder muscles receive a richer supply of descending fibers from the ipsilateral hemisphere than those of the elbow and wrist [2,27,32]. The effectiveness of connectivity for the ipsilateral corticospinal tract in human was similar to that in the monkey. Ipsilateral motor-evoked potentials can be elicited regularly in proximal muscles with transmagnetic stimulation, but it is more difficult to obtain a similar response from the distal muscle of the upper extremity [13,42].

Interestingly, a larger proportion of GS was attributable to irradiation from synergists of extension and adduction than it was for those of flexion and abduction. Perhaps, the unique GS spatial representation reflects a need of biomechanical constraint for postural stabilization [18,21] when a subject performed a strenuous contraction in the testing chair. However, from a biomechanical standpoint, the enhancement of the RE of the ECR by contralateral wrist movement was not expected. Such an irradiation outcome in the ECR might be a case of "cortical activation theory" as proposed by Todor and Lazarus [39]. According to the theory, preferential use of the muscles for supportive purpose allows some modification of cortical plasticity and results in a greater irradiation to the muscles of the opposite limb [38]. Since the ECR typically serves as a supporter of the wrist joint during writing, typing, and many functional activities undertaken in daily life [19,36], use-dependent irradiation of the ECR during homolateral wrist movement is understandable. Our EMG finding concurs, conceptually, with the results of the study of Armatas et al. [5] who reported that a greater level of mirrored movement of the hand was recorded for finger movement of supporting roles (e.g., the small finger) as compared to that for manipulative purposes (e.g., the index finger).

4.3. Methodological concerns

Before jumping into conclusion, methodological concerns and limitations of this study should be addressed, particularly for normalization of the EMG to obtain GS in both limb muscles. The rationales in favor for this normalization are based on (1) a consistent background activity of all muscles at rest between two limbs, and (2) EMG amplitude of homologous muscles between the two sides is comparable. Several researchers reported no statistical difference in motor unit action potential and M wave between the dominant and non-dominant limbs [7,8], and maximal amplitude of surface EMG recorded in both limb muscles was similar [26,41]. Consequently, we believe that laterality of GS intensity and symmetry of GS spatial representation in this study was not due to physical factors associated with recording techniques, such as skin fold thickness and impedance and so on. In fact, we consider expressing irradiated activity with the SNE value rather than as a percentage of maximal voluntary EMG, because the maximal torque of a target movement can hardly be achieved by a single muscle. For instance, the development of elbow flexion requires coactivation of several functional synergists (brachialis, biceps brachii, and brachioradialis), whereas relative contributions from these muscles to the maximal flexion torque vary with the position of the elbow. Traditional normalization with percentage of maximal voluntary EMG is not physiologically appropriate. On the other hand, SNE allows us to realize the multiples of net irradiation level relative to relaxed background activity and changes in GS spatial pattern with ease, accounting for the contribution of undersized irradiation in some muscles [22]. However, the interpretation of GS characteristics based on SNE and RE should be performed with care, granting that SNE and RE may not provide sufficient information to compare the real excitation level among non-homologous muscles. It may well to say that irradiated EMG activity of the PEC contributed to more GS than any other muscles in the upper limb for both the dominant condition and non-dominant conditions. Since irradiated activities of the recorded muscles were not normalized to their maximal values, whether the PEC was facilitated by the contralateral movements to the largest extent remains uncertain.

5. Conclusion

Global synkinesis typically involves with coactivation of several non-homologous muscles with greater GS intensity when maximal isometric contraction of the non-dominant upper limb was performed. For a given joint of the non-dominant limb, GS intensity was dependent strongly on movement direction, but this directional effect was not equally manifest for the dominant condition. For either the non-dominant or dominant condition, relative contributions of individual irradiated muscles to contralateral GS was largely resembled subjective to a similar joint effect. Linking our observation to known physiological evidence, it seems likely that use-dependent cortical adaptation and topographical representation of the ipsilateral pathways gave rise to structured GS in this study.

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